

Biasing:

Start with bias trimmer R8 set to minimum resistance.
Power up with no load first; transformer disconnected,
input shorted to ground, and current limiting if available.

Measure DC millivolts across R1 and R2.
Because R1 and R2 are 1Ω, 1mV = 1mA.

Aim for 50mV–80mV across each resistor.
Let the amp warm up for 10–15 minutes.
Measure again and adjust if needed.
Start with minimum, 50mV. Then after
10–15 of warming up adjust further to the
target value of your choice, somewhere
between 50mV and 80mV.
If minimum R8 already gives excessive idle current,
power down immediately; do not continue adjusting.
R1/R2 currents should settle reasonably close to
each other. Large mismatch may indicate device/wiring problems.

Negative feedback (NFB pot):

Counter-clockwise = LOOSE,
maximum feedback resistance,
(slightly above 1M, R18 + R19 + NFB pot)
least negative feedback,
lowest damping.

Clockwise = TIGHT,
minimum feedback resistance
(44k, the sum of R18 and R19),
maximum negative feedback,
highest damping.

Nominal midband input impedance is around 33k.
You need a preamp/buffer in front of the amp
that can supply a line-level low impedance signal.

Put R4 and R5 as close to the gate pins as possible.
Also put the zener diodes D2, D3, D4, and D5 as close
to the MOSFET pins as possible.

Attach Qbias to the same heatsink as Q–Nch and Q–Pch.
Put it between them; it needs thermal tracking.
Qvas does not need to be attached to the heatsink.

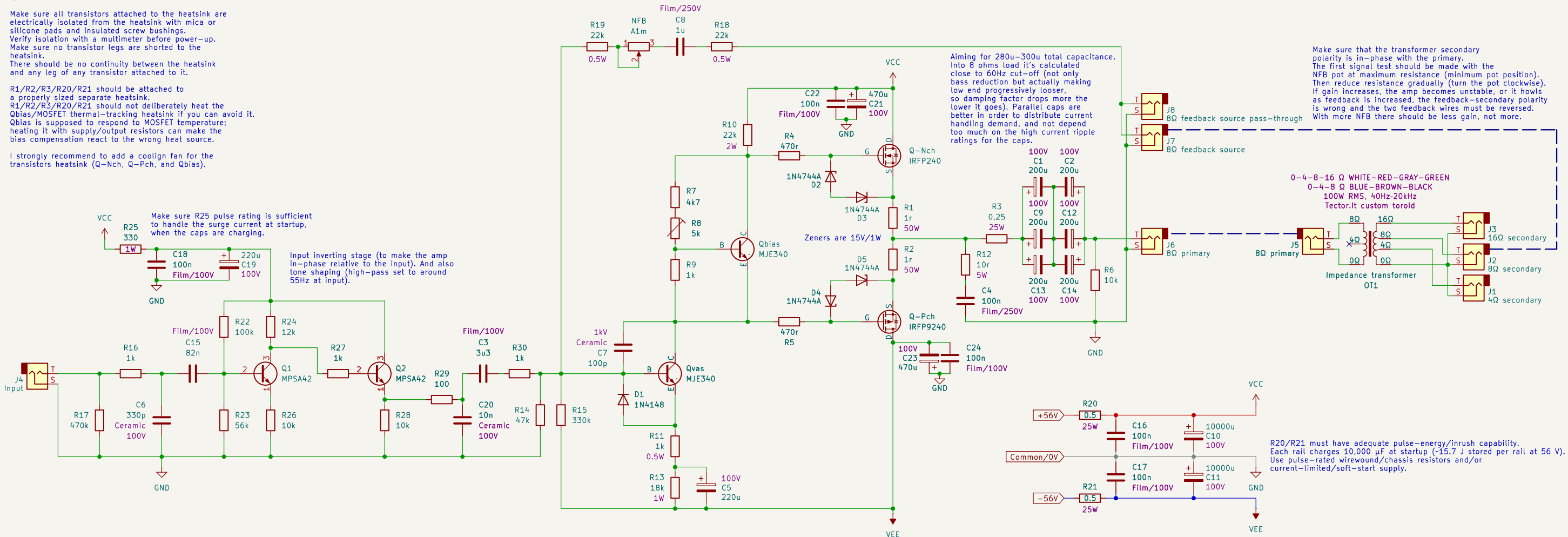
Make sure all transistors attached to the heatsink are
electrically isolated from the heatsink with mica or
silicone pads and insulated screw bushings.
Verify isolation with a multimeter before power-up.
Make sure no transistor legs are shorted to the
heatsink.
There should be no continuity between the heatsink
and any leg of any transistor attached to it.

R1/R2/R3/R20/R21 should be attached to
a properly sized separate heatsink.
R1/R2/R3/R20/R21 should not deliberately heat the
Qbias/MOSFET thermal-tracking heatsink if you can avoid it.
Qbias is supposed to respond to MOSFET temperature;
heating it with supply/output resistors can make the
bias compensation react to the wrong heat source.

I strongly recommend to add a coolign fan for the
transistors heatsink (Q–Nch, Q–Pch, and Qbias).

LF VOICING:

–55 Hz input HPF + –57–60 Hz passive output–coupling pole are cascaded,
so total electrical LF rolloff starts higher than either pole alone.
Exact response varies somewhat with NFB setting.
Despite this, LF damping falls strongly with frequency because of the
series output capacitor. Speaker/cab resonances therefore have more
influence, so perceived bass can remain strong and "bloomin'" while
unwanted sub–bass is reduced.



WORK IN PROGRESS...

Sheet: /
File: wenzels-powered-guitar-cabinet-2606-single-amp.kicad_sch

Title: Wenzel's Powered Guitar Cabinet 2606

Size: A3 Date: October 2026

Rev: r1-wip-4

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